Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For

AI Vibe Coding with Python

Course Code: C138

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version 1.0

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DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
1.0	30 September 2026	Initial release — 5 topics, 20 labs, CardGuard fraud-screening application, delivered over 2 days (9:30am – 5:30pm, 7.5 instructional hours per day).	Dr. Alfred Ang

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Course Information

Course Title	AI Vibe Coding with Python
Course Reference	C138
Training Provider	Tertiary Infotech Academy Pte Ltd (UEN 201200696W)
Duration	2 days · 7.5 instructional hours per day (15 hours)
Daily Timing	9:30 am – 5:30 pm (30-minute lunch break)
Mode	Instructor-led, hands-on labs using uv, an AI coding assistant, VS Code and Docker
Trainer	Dr. Alfred Ang

Learning Outcomes

On completion of this course, learners will be able to:

- LO1: Apply vibe coding practices — set up a reproducible Python project with uv, and use an AI coding assistant to scaffold, explain and refactor working code.
- LO2: Design and implement object-oriented Python — classes, encapsulation, inheritance and composition — using AI-assisted conversational design.
- LO3: Build data analytics pipelines with pandas — load, clean, transform, group and aggregate realistic datasets into reportable results.
- LO4: Model and validate data with Pydantic and expose it through a FastAPI application with typed request and response models.
- LO5: Package and deploy a Python application — dependency locking, configuration, containerisation and a running deployed service.

Course Schedule

Day 1 — Vibe Coding Foundations, Object-Oriented Python & pandas

Time	Duration	Topic / Activity
9:30–9:45	15 min	Welcome, course introduction and ground rules
9:45–10:15	30 min	Topic 1 — Vibe Coding Foundations: uv Projects · AI Coding Assistants · Prompting Patterns · AI-Assisted Refactoring (concepts + demo)
10:15–12:30	135 min	Hands-on: Lab 1: Set Up a Reproducible Python Project with uv; Lab 2: Write Effective Prompts for an AI Coding Assistant; Lab 3: Review and Correct AI-Generated Code; Lab 4: Refactor Working Code Conversationally
12:30–13:00	30 min	Lunch break
13:00–13:30	30 min	Topic 2 — Object-Oriented Programming in Python:

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		Classes · Encapsulation · Inheritance · Composition · Dunder Methods (concepts + demo)
13:30–15:35	125 min	Hands-on: Lab 5: Model a Domain with Classes and Dunder Methods; Lab 6: Encapsulate State with Properties and Validation; Lab 7: Build a Rule Hierarchy with Inheritance and Polymorphism; Lab 8: Compose Rules into an Engine
15:35–16:05	30 min	Topic 3 — Data Analytics with pandas: DataFrames · Cleaning · Transformation · GroupBy · Aggregation · Reporting (concepts + demo)
16:05–17:15	70 min	Hands-on: Lab 9: Load and Explore Transactions with pandas; Lab 10: Handle Errors with Exceptions
17:15–17:30	15 min	Day 1 recap and Q&A

Total instructional time: 450 minutes (7.5 hours), 9:30am – 5:30pm, with a 30-minute lunch break.

Day 2 — pandas, Pydantic & FastAPI, Packaging & Deployment

Time	Duration	Topic / Activity
9:30–9:45	15 min	Day 1 recap
9:45–11:35	110 min	Hands-on: Lab 11: Build Per-Cardholder Baselines with GroupBy; Lab 12: Detect Velocity Bursts with Rolling Time Windows
11:35–12:35	60 min	Topic 4 — Data Modelling with Pydantic and FastAPI: Typed Models · Validation · Endpoints · Request/Response Schemas · Auto Docs (concepts). Hands-on: Lab 13: Replace Hand-Written Validation with Pydantic Models
12:35–13:05	30 min	Lunch break
13:05–15:00	115 min	Hands-on: Lab 14: Expose the Screening Engine as a FastAPI Endpoint; Lab 15: Persist Screening Decisions to SQLite; Lab 16: Test the API with pytest and httpx
15:00–15:30	30 min	Topic 5 — Packaging and Deployment: uv Lockfiles · Configuration · Containers · Deploying a Service (concepts + demo)
15:30–17:15	105 min	Hands-on: Lab 17: Build a Fraud Analyst Dashboard with Streamlit; Lab 18: Externalise Configuration and Protect

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		Secrets; Lab 19: Lock Dependencies and Containerise the API; Lab 20: Deploy the Full Stack with Docker Compose
17:15–17:30	15 min	Course wrap-up, Q&A and course feedback

Total instructional time: 450 minutes (7.5 hours), 9:30am – 5:30pm, with a 30-minute lunch break.

Lab Reference

Topic	Weighting	Labs
Topic 01: Vibe Coding Foundations	15–20%	Lab 1, Lab 2, Lab 3, Lab 4
Topic 02: Object-Oriented Programming in Python	20–25%	Lab 5, Lab 6, Lab 7, Lab 8
Topic 03: Data Analytics with pandas	20–25%	Lab 9, Lab 10, Lab 11, Lab 12
Topic 04: Data Modelling with Pydantic and FastAPI	20–25%	Lab 13, Lab 14, Lab 15, Lab 16
Topic 05: Packaging and Deployment	15–20%	Lab 17, Lab 18, Lab 19, Lab 20

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